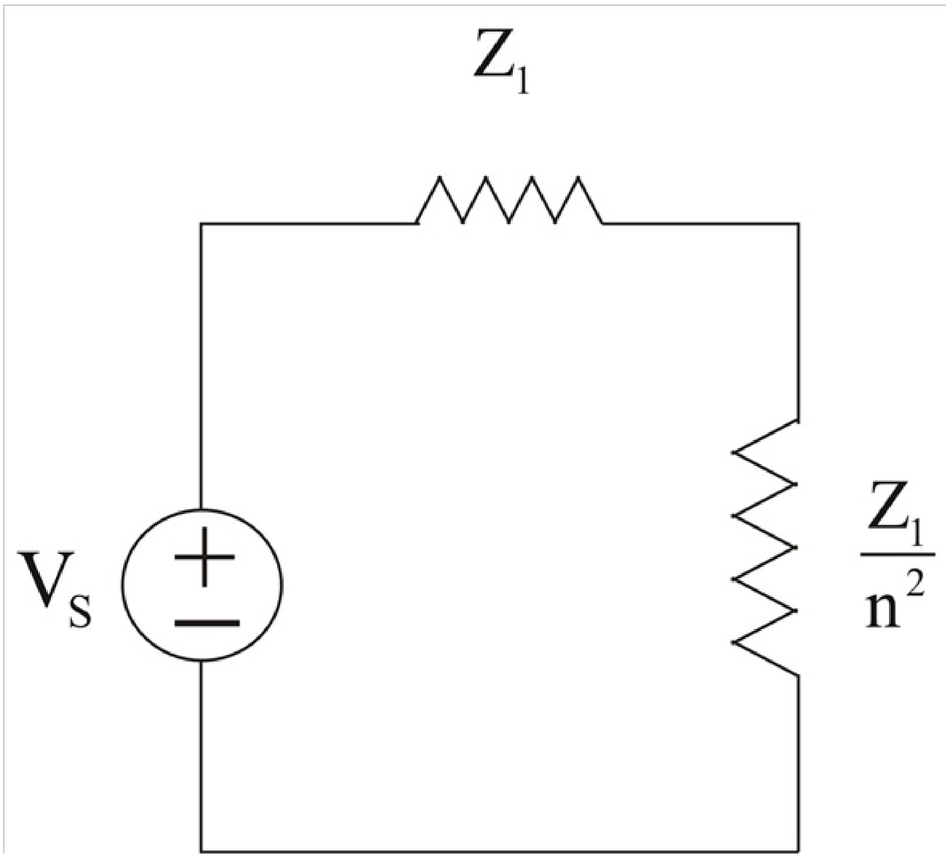
Chapter 13, Problem 85P

Problem

A stereo amplifier circuit with an output impedance of 7.2 kΩ is to be matched to a speaker with an input impedance of 8 Ω by a transformer whose primary side has 3,000 turns. Calculate the number of turns required on the secondary side.

Step-by-step solution

Step 1 of 2



Step 2 of 2

For maximum power transfer,

$$Z_1 = \frac{Z_L}{n^2}$$

$$\Rightarrow \quad n^2 = \frac{Z_L}{Z_1} = \frac{8}{7200}$$

$$n^2 = \frac{1}{900}$$

$$n = \frac{1}{30} = \frac{N_2}{N_1}$$

Thus,

$$N_2 = \frac{N_1}{30} = \frac{3000}{30}$$

$$= 100 \; Turns \; .$$